Google Cloud H2S

Gen AI Academy

APAC Edition

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PARTICIPANT DETAILS

a. Participant name : **Hariharan C**

b. Problem Statement : **Build and deploy a production-ready AI agent using Gemini and ADK that performs a single task via an HTTP endpoint on Cloud Run.**

c. Github repo link : ***https://github.com/BlackWizard123/Track-1---Financial_Risk_Classifier_Agent***

d. Cloud Run URL : ***https://financial-risk-agent-1059652519537.asia-south1.run.app/***

Brief about the idea

We built an **AI-powered Financial Risk Classifier Agent** that analyzes a user's financial situation from natural language input.

- Takes unstructured financial text (income, expenses, debts)
- Uses **Gemini** to understand and evaluate financial signals
- Classifies risk into **Low / Medium / High**
- Returns **structured insights** with:
    - Reasoning
    - Summary
    - Improvement suggestions
    - Stability plan

Deployed as a **scalable API on Cloud Run**, making it easily accessible and production-ready.

Meeting the Build Criteria

How did you navigate the requirements of your chosen track to build a custom solution - using ADK, MCP, or AlloyDB AI that effectively addresses the 'What You Must Build' criteria?

- Implemented a **single AI agent** focused on financial risk classification
- Integrated **ADK structure** to align with agent-based architecture
- Designed a clear **HTTP endpoint (/analyze)** using FastAPI
- Ensured **input → processing → structured output** flow
- Used **Gemini model** for intelligent analysis and reasoning
- Deployed on **Cloud Run** for scalability and accessibility

✓ Fully satisfies the requirement of building a **production-ready, callable AI agent**

Opportunities

- **How different is it from any of the other existing ideas?**

Uses natural language input instead of structured financial data, making it more intuitive than traditional tools

- **How will it be able to solve the problem?**

Translates user financial descriptions into clear risk insights and actionable recommendations

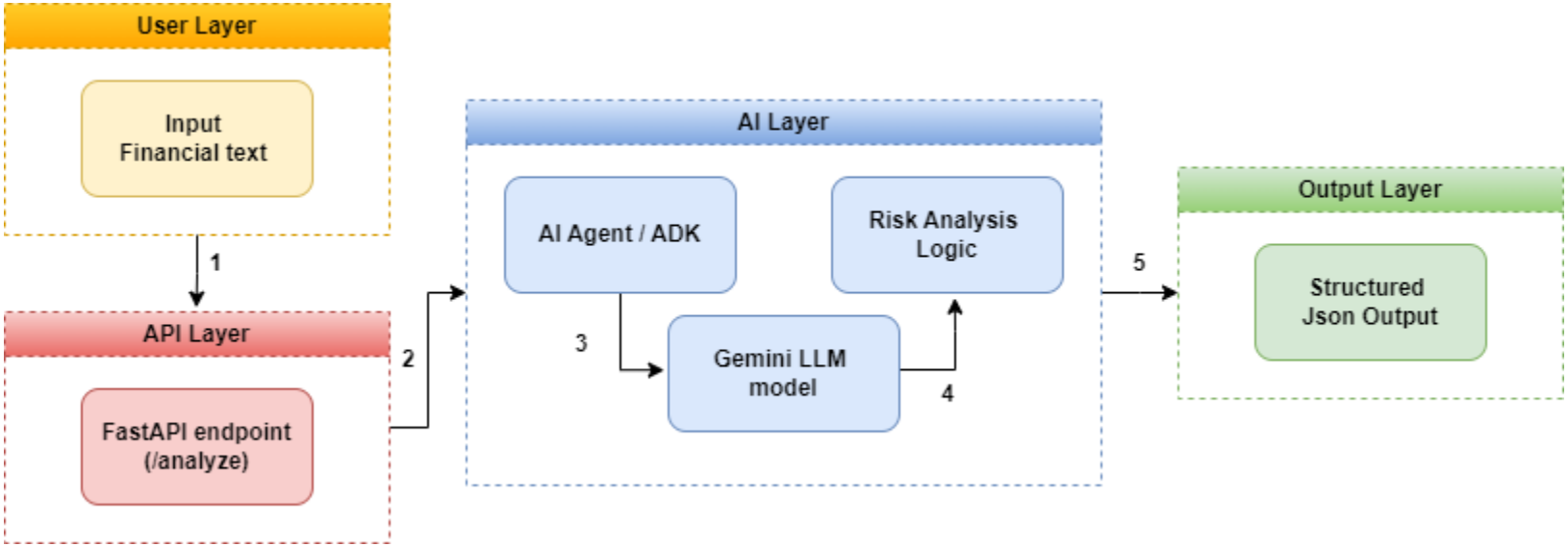
- **USP of the proposed solution**

Translates user financial descriptions into clear risk insights and actionable recommendations

List of features offered by the solution

- AI-powered financial risk classification (Low / Medium / High)
- Accepts natural language financial input
- Structured JSON output (not just plain text)
- Clear reasoning behind risk classification
- Actionable improvement suggestions
- Stability planning guidance
- FastAPI-based HTTP endpoint (/analyze)
- Deployed on Cloud Run (scalable & accessible)
- ADK-based agent structure
- Optional UI for interactive demo

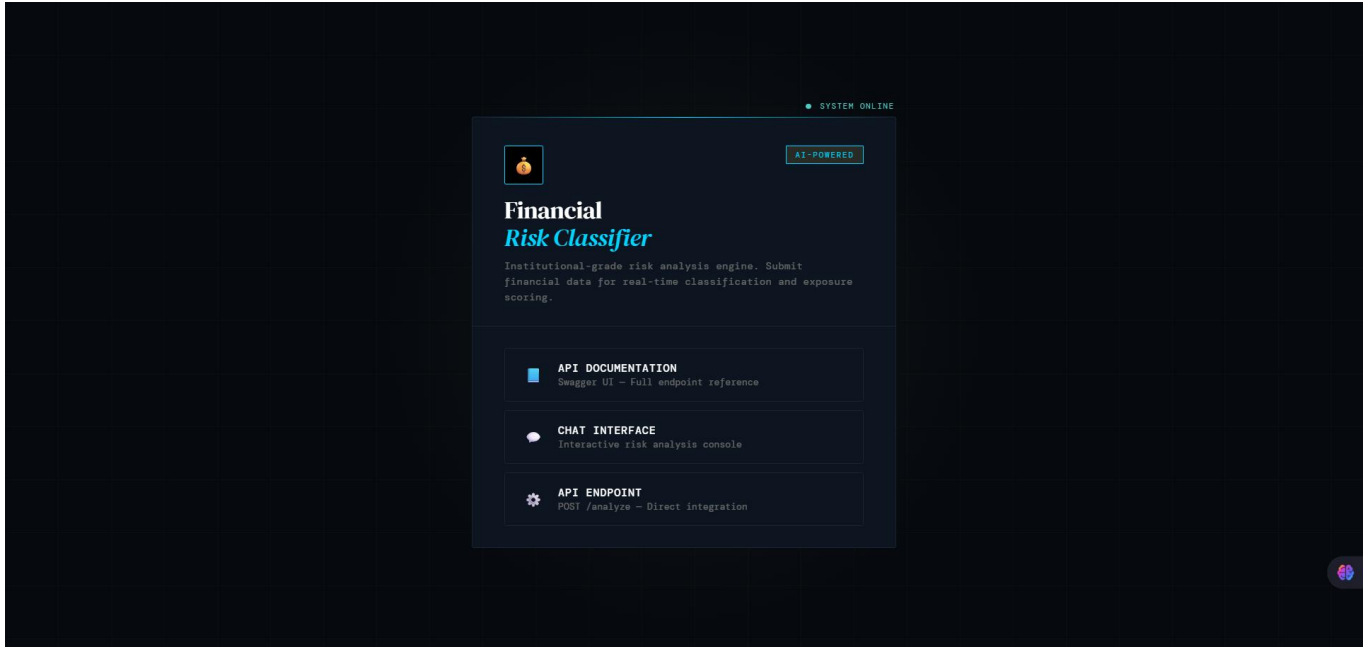
Process flow diagram or Use-case diagram



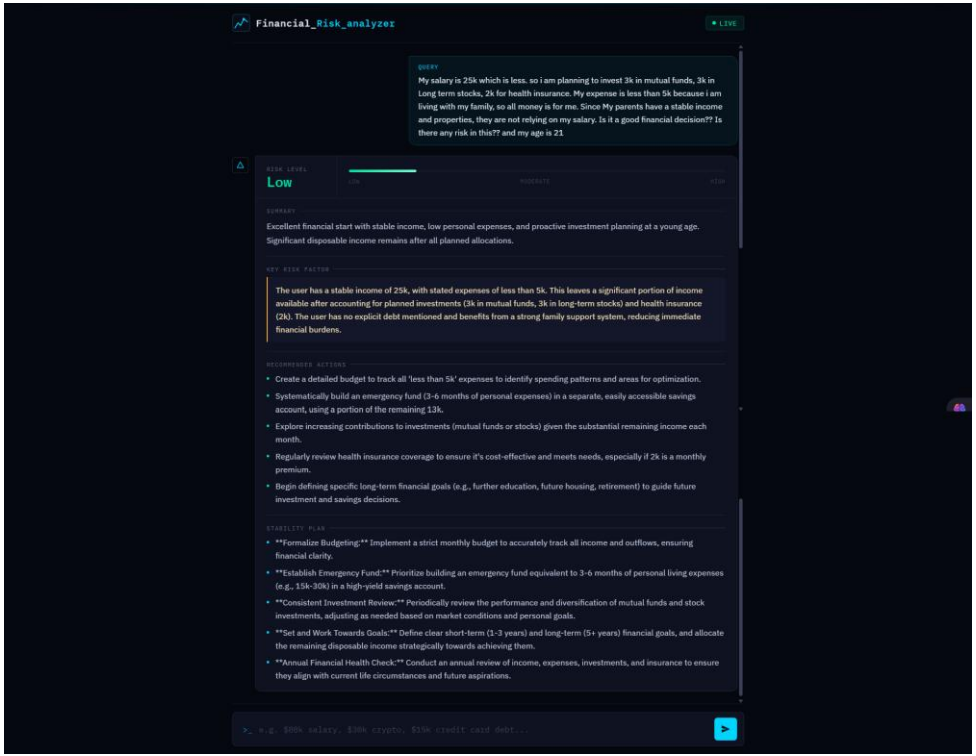
Technologies used in the solution

- **Python** – Core programming language
- **FastAPI** – Backend API framework
- **Google Gemini** – AI model for analysis and reasoning
- **Docker** – Containerization
- **Google Cloud Run** – Serverless deployment
- **Google ADK (Agent Development Kit)** – Agent structure and orchestration
- **HTML/CSS (Optional UI)** – Simple user interface for demo

Snapshots of the prototype



https://financial-risk-agent-1059652519537.asia-south1.run.app/



https://financial-risk-agent-1059652519537.asia-south1.run.app/ui

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Thank you